13. Generate a Parse Tree Using a Context-Free Grammar

Aim: To generate a parse tree for a given sentence using a context-free grammar in Python.

Procedure:

1. Import NLTK.
2. Define grammar production rules.
3. Create a chart parser.
4. Provide the input sentence.
5. Display the parse tree.

Code:

```
import nltk
grammar = nltk.CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V
Det -> 'the'
N -> 'cat'
V -> 'sleeps'
""")
parser = nltk.ChartParser(grammar)
sentence = "the cat sleeps".split()
for tree in parser.parse(sentence):
    print(tree)
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/13.py"
(S (NP (Det the) (N cat)) (VP (V sleeps)))
```

Result: A parse tree was successfully generated for the given sentence.